ARCHES: A Randomized, Phase III Study of Androgen Deprivation Therapy with Enzalutamide or Placebo in Men with Metastatic Hormone-Sensitive Prostate Cancer

Andrew J. Armstrong, et al

APPENDIX

List of Investigators for the ARCHES Study

The following investigators, listed in alphabetical order by country, participated in the ARCHES study: **Argentina –** M. Brown Arnold (Rosario), S. Metrebian (Córdoba), E. Staneloni (Buenos Aires), J. Zarba (San Miguel de Tucumán); **Australia –** E. Abdi (Tweeds Head), A. Azad (Clayton), T. Bonaventura (Waratah), A. Boyce (Lismore), S. Brown (Ballarat), N. Corcoran (Parkville), P. Craft (Garran), A. Guminski (St. Leonards), H. Gurney (Macquarie University), L. Horvath (Camperdown), G. Kichenadasse (Adelaide), E. Kwan (Clayton), W. Patterson (Woodville South), K. Pittman (Woodville South), G. Van Hazel (Perth), S. Wong (St. Albans); **Belgium –** F. Ameye (Gent), L. D'Hondt (Yvoir), L. Goeman (Roeselare), K. Lesage (Kortrijk), G. Pelgrims (Turnhout), V. Richard (Mons), D. Waltregny (Liège); **Canada –** R. Casey (Oakville), R. Egerdie (Kitchener), C. Ferrario (Montreal), D. Finch (Kelowna), N. Fleshner (Toronto), J. Giddens (Brampton), T. Kinahan (Kelowna), L. Klotz (Toronto), S. North (Edmonton), P. Ouellette (Granby), R. Siemens (Kingston), J. Simard (Laval), G. Vrabec (Abbotsford), J. Zadra (Barrie); **Chile –** A. Acevedo Gaete (Viña Del Mar), C.L. Caglevic Medina (Santiago), P. Gonzalez Mella (Reñaca), J.L. Leal (Santiago), M. Mahave Caceres (Santiago), F. Orlandi Jorquera (Providencia), A. Salazar (Santiago), E. Yañez Ruiz (Temuco); **Denmark –** J.S. Alban Veloso (Herlev), M. Borre (Aarhus N), K. Brasso (Copenhagen), J.B. Jensen (Holstebro), J. Johansen (Holstebro), N. Langkilde (Aalborg), M. Poulsen (Odense C), P. Rathenborg (Herlev); **Finland –** P. Boström (Turku), T. Joutsi (Pori), A. Kaipia (Pori), T. Marttila (Seinäjoki), C. Palmberg (Pietarsaari), H. Ronkainen (Oulu), K. Taari (Helsinki), T. Tammela (Tampere); **France –** A.-R. Azzouzi (Angers), M. Colombel (Lyon Cedex 03), A. de la Taille (Créteil), A. Fléchon (Lyon Cedex 09), C. Helissey (Saint-Mande), N. Houede-Tchen (Nimes), F. Joly (Caen), F. Priou (La Roche sur Yon), G. Robert (Bordeaux), A. Ruffion (Pierre

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Trial Registration

The trial was submitted to ClinicalTrials.gov on February 5, 2016 and first posted on February 9, 2016. The first patient was screened and consented on March 7, 2016, this date is noted as the actual study start date on ClinicalTrials.gov. The first patient that was randomized was screened and consented on March 18, 2016, and then randomized on March 21, 2016.

Supplementary Methods

Inclusion and Exclusion Criteria

Eligible patients maintained androgen deprivation therapy (ADT) with a luteinizing hormone-releasing hormone (LHRH) agonist/receptor antagonist during study treatment or had a history of bilateral orchiectomy, had no severe concurrent diseases, comorbidities, or infections, suspected brain metastases or active leptomeningeal disease, history of another invasive cancer within 3 years of screening (with the exception of fully treated cancers with a remote probability of recurrence based on investigator assessment), history of seizure or a condition that may confer a predisposition to seizure, history of loss of consciousness or transient ischemic attack within 12 months prior to day 1, hypersensitivity reaction to the active pharmaceutical ingredient or any of the study capsule components, clinically significant cardiovascular disease, or gastrointestinal disorder affecting absorption.

Patients were excluded if they had received previous pharmacotherapy, radiation therapy, or surgery for metastatic prostate cancer, with the exception of:

- ≤ 3 months, or ≤ 6 months (if the patient was treated with docetaxel), of ADT with LHRH analogs or orchiectomy, with or without concurrent antiandrogens prior to day 1, with no radiographic evidence of disease progression or rising prostate-specific antigen (PSA) levels prior to day 1;
- One course of palliative radiation or surgical therapy to treat symptoms resulting from metastatic disease administered at least 4 weeks prior to day 1;
- ≤ 6 cycles of docetaxel therapy completed within 2 months of day 1 and no evidence of disease progression during or after completion;

- Prior ADT given for < 39 months in duration and > 9 months before randomization as neoadjuvant/adjuvant therapy.

Eligible patients had not received prior aminoglutethimide, ketoconazole, abiraterone acetate, or enzalutamide for the treatment of prostate cancer, or any investigational agent that inhibits the androgen receptor or androgen synthesis. Patients who received treatment with 5-α reductase inhibitors (finasteride, dutasteride), estrogens, cyprotoerone acetate, androgens, systemic glucocorticoids (greater than the equivalent of 10 mg per day of prednisone), herbal medications that have known hormonal antiprostate cancer activity and/or are known to decrease PSA levels, investigational agents, or who had major surgery, within 4 weeks prior to day 1, or who received bisphosphonates or denosumab within 2 weeks prior to day 1, unless administered at stable dose or to treat diagnosed osteoporosis, were excluded.

Concomitant Treatment

Maintenance of concomitant treatment with ADT, either bilateral orchiectomy or a LHRH agonist/receptor antagonist, during study treatment, was required as per standard of care (SOC).

Concomitant treatment with 5-α reductase inhibitors (finasteride, dutasteride), estrogens, cyprotoerone acetate, biologic or other agents with antitumor activity against prostate cancer (other than the exceptions listed in the above section) systemic glucocorticoids (greater than the equivalent of 10 mg per day of prednisone), herbal medications that have known hormonal antiprostate cancer activity and/or are known to decrease PSA levels, androgens, investigational agents, and bisphosphonates or denosumab, unless stabilized for 2 weeks prior to randomization and held constant, as tolerated, throughout study treatment or administered for diagnosis of osteoporosis, was prohibited.

Permitted concomitant therapies included, but were not limited to, the following treatments:

- Blood transfusions and growth factor support as per SOC and institutional guidelines;
- Steroid use (for indications other than prostate cancer) as per SOC;
- Pain therapy as per SOC and institutional guidelines;
- Palliative radiation therapy including external beam radiotherapy or systemic radionuclides (e.g. Samarium or Strontium);
- Vaccine therapy that has prior market authorization and is not intended to treat prostate cancer;
- Palliative surgical procedures to treat skeletal-related events;
- Hormonal treatment for treating complications of LHRH analogue treatment (e.g, hot flashes) was allowed with Medical Monitor approval;
- Flutamide, bicalutamide, or nilutamide, if given concurrently with a LHRH agonist or receptor antagonist to prevent flare.

Radiographic Progression-Free Survival (rPFS) and Overall Survival (OS): Original Statistical Analyses

The final rPFS analysis was originally planned to be conducted after 510 rPFS events to detect a hazard ratio (HR) of 0.75 with 90% power, based on a two-sided log-rank test and 5% significance level. The final OS analysis was to be conducted after 450 deaths. The minimum number of rPFS events was amended to 262 to detect a HR of 0.67 with 90% power, assuming a median rPFS for the placebo arm of 20 months, based on recent data in men with mHSPC.[1] The final OS analysis was amended accordingly and will be performed when approximately 342 deaths have occurred to provide 80% power to detect an HR of 0.73 at a 4% significance level.

An interim OS analysis at the time of the final rPFS analysis was added at a significance level calculated using the O'Brien-Fleming function to control the overall alpha. Amendments to the rPFS and OS statistical analyses plans were performed prior to unblinding.

Multiplicity Adjustment Strategy

All secondary end point analyses were performed at the time of the rPFS final analysis. As the primary end point statistical analysis test (conducted at a two-sided significance level of 0.05) was statistically significant, the following six key secondary end points were tested: time to PSA progression, time to initiation of new antineoplastic therapy, the rate of PSA decline to < 0.2 ng/mL (PSA undetectable rate), objective response rate, the time to deterioration in urinary symptoms from the Quality of Life PRostate-Specific questionnaire 25, and OS. To maintain the family-wise two-sided type I error rate at 0.05, a parallel testing strategy between OS (with allocated type I error rate 0.04) and the other five end points (with allocated type I error rate 0.01) was performed, as summarized in Fig A1. The O'Brien-Fleming alpha spending function was used to determine the stopping boundaries based on the number of events observed at the interim lock to control the overall two-sided alpha for the final analysis of OS at 0.04, as time to deterioration in urinary symptoms was not significant.

Adjusting for Missing Data

As a general principle, no imputation of missing data was performed. Exceptions included the start and stop dates of adverse events (AEs), previous and concomitant medications, the date of initial diagnosis (to estimate the relative study day to calculate cancer duration), dates of cancer treatment (e.g. previous procedure, previous radiotherapy, etc.), the last dose date, and the date of death. The imputed dates were used to allocate the relative study day and, in addition, to determine whether an AE was/was not treatment emergent. Cases where the onset date of an AE was (partially) missing were addressed during the data review meeting in order to determine whether the AE was considered treatment emergent or not.

Imputation on missing non-prostate-cancer-related medication dates (to categorize them as previous medications or concomitant medications, or post-treatment) and AE dates (to categorize them as treatment emergent AEs or not) was performed as follows:

- Incomplete start day from start date and the corresponding end date was complete: used the later of (first day of the month, first dosing day if first dosing month); but if later than the end date, then imputed the start day as the day of the end date.
- Incomplete start day from start date and incomplete End Day from end date: used the later of (first day of the month, first dosing day if first dosing month).
- Incomplete end day from end date: used the earliest of (last day of the month, day of the 30-day follow-up visit if it was the month of the 30-day follow-up visit).
- Incomplete Month or Year: no imputation.

Imputation on missing date of initial diagnosis (cancer duration) and prior cancer treatment, including prostate cancer drug medication/therapies, (e.g. start date, stop date, or date of procedure) was performed as follows:

- Incomplete Day: used the 15th day of the month, if month/year was before first dosing or after last dosing (for start date imputation, but if later than the end date, then imputed the start day as the day of the end date; for end date imputation, but if earlier than the start date, then imputed the end day as the day of the start date).

- Incomplete Month: used 1st July if the Year was before Year of first dosing, otherwise missing.
- Incomplete Year: no imputation, the derived variable was considered to be missing.

If missing for patients who started treatment, the last dose date of treatment was imputed as follows:

- Incomplete Day only: used the earliest of (last day of the month, end of treatment [form] day, if on the same month and year, day of the 30-day follow-up visit, if on the same month and year).
- If fully missing or Incomplete Month or/and Year: the date was imputed by the earliest of (end of treatment [form] date, date of the 30-day follow-up visit).

If partially missing, the date of death was imputed as follows:

- Incomplete Day: used the earliest of (last day of the month, end of study [form] day).
- Incomplete Month or Year: no imputation.

Imputation methods were not used to determine other end points.

Table A1. End Point Definitions and Analysis Method.

End point variable	Definition and analysis method
Primary	
rPFS	Time from randomization to the first objective evidence of radiographic disease progression as assessed by central review or death (defined as death from any cause within 24 weeks from study drug discontinuation), whichever occurs first
	Patients who discontinued study treatment without confirmed radiographic progression continued the radiographic assessment schedule until radiographic progression was confirmed or the number of radiographic progression events for the final rPFS analysis was reached
	Radiographic disease progression is defined as progressive disease by RECIST, version 1.1,[2] for soft tissue disease or by appearance of two or more new lesions on bone scan (see Supplementary Table S2)
	Efficacy analyses were conducted on the intent-to-treat population, defined as all randomized patients
	A two-sided stratified log-rank test at a level of significance of 0.05 was used to compare rPFS between treatment groups. Stratification factors were the prior use of docetaxel and the volume of disease. The median was estimated using the Kaplan-Meier method with its 95% CI calculated using the Brookmeyer and Crowley method. Treatment benefit was summarized by a single HR with its 95% CI based on the Cox proportional hazards model, stratified for disease volume and prior use of docetaxel therapy
	Prespecified subgroup analysis of rPFS was performed to determine whether the treatment effect was concordant among subgroups. To avoid possible issues related to the small number of events, subgroup analyses were not adjusted for the stratification factors
	To assess the robustness of the primary analysis of rPFS, the following sensitivity analyses were performed:

	Sensitivity analysis of rPFS for impact of all deaths (with no time limit) as events, defined as time from randomization to the first objective evidence of radiographic disease progression as assessed by central review or death (defined as death from any cause at any time), whichever occurs first Sensitivity analysis of rPFS for impact of radiographic progression documented by central review according to PCWG2 criteria,[3] defined as time from randomization to the first objective evidence of radiographic disease progression as assessed by central review (defined as death from any cause within 24 weeks from study drug discontinuation), whichever occurs first Radiographic disease progression is documented according to PCWG2 criteria,[3] more specifically, the documentation and confirmation required for the determination of radiographic disease progression on bone lesions as described in Table S2, with the exception that the appearance of ≥ 2 new bone lesions is to be compared to baseline (or week 13) bone scan for week 25 or later visits
Key secondary	
Time to PSA progression	Time from randomization to a ≥ 25% increase and an absolute increase of ≥ 2 ng/mL above the nadir (i.e. lowest PSA value observed postbaseline or at baseline), which is confirmed by a second consecutive value at least 3 weeks later. Only PSA assessments taken prior to the initiation of new antineoplastic therapy were evaluated The analysis method was the same as the primary analysis for rPFS with a significance level of 0.01
Time to initiation of new antineoplastic therapy	Time from randomization to the initiation of antineoplastic therapy (including cytotoxic and hormonal therapies) subsequent to the study treatments The analysis method was the same as the primary analysis for rPFS with a significance level of 0.01
PSA undetectable rate	The percentage of patients with detectable (≥ 0.2 ng/mL) PSA at baseline, which becomes

	undetectable (< 0.2 ng/mL) during study treatment. Only PSA assessments taken prior to the initiation of new antineoplastic therapy were evaluated A stratified Cochran-Mantel-Haenszel score test was used to compare the response rates between treatment groups with a significance level of 0.01
ORR	The percentage of patients with measureable disease at baseline who achieved a complete or partial response in their soft tissue disease using the RECIST, version 1.1, criteria[2] (see Table A2) The analysis method was the same as for PSA undetectable rate
Time to deterioration in urinary symptoms	Time from randomization to the first deterioration in urinary symptoms, defined as an increase in urinary symptoms scores, using a modified urinary symptoms scale derived from a selected subset of symptoms from the QLQ-PR25 questionnaire module[4] (including three items, Q31–Q33) by $\geq$ 50% of the standard deviation observed in the modified urinary symptoms scale at baseline The analysis method was the same as the primary analysis for rPFS with a significance level of 0.01
OS	Time from randomization to death from any cause All patients will be followed for survival until death, loss to follow-up, withdrawal of consent, or study termination by the sponsor The O'Brien-Fleming alpha spending function was used to account for the interim analysis of OS, which was performed at the time of the final rPFS analysis The analysis method was the same as the primary analysis for rPFS The final analysis of OS will be tested at 0.04, as time to deterioration in urinary symptoms was not significant
Secondary	
Time to first SSE	Time from randomization to the occurrence of the first SSE, defined as radiation or surgery to bone, clinically apparent pathologic bone fracture, or spinal cord compression

	The analysis method was the same as the primary analysis for rPFS
Time to castration resistance	Time from randomization to the first castration-resistant event (radiographic disease progression, PSA progression, or SSE with castrate levels of testosterone [< 50 ng/dL]), whichever occurs first The analysis method was the same as the primary analysis for rPFS
Time to deterioration of QoL	Time from randomization to a 10-point reduction of the FACT-P total score, which is the combined score from 27 core items that assess subject function in four domains: physical, social/family, emotional, and functional well-being, in addition to 12 site-specific items which assess for prostate-related symptoms, each rated on a 0 to 4 Likert-type scale, in this instrument designed specifically for use with prostate cancer. Higher scores represent better QoL (range 0–156)[5] The analysis method was the same as the primary analysis for rPFS
Time to pain progression	Time from randomization to an increase of ≥ 30% in pain severity score from baseline using the BPI-SF, a validated instrument that is a subject self-rating scale (ranging from 0 to 10) assessing level of pain, effect of the pain on activities of daily living, and analgesic use The analysis method was the same as the primary analysis for rPFS
Safety	Safety analyses were conducted on the safety population, defined as all randomized patients who received ≥ 1 dose of the study drug Frequency and severity of adverse events, safety laboratory tests, physical examinations, electrocardiograms and vital signs were summarized descriptively
Additional prespecified analyses (per PRO statistical analysis plan)	
QoL over time	Mean FACT-P total score over time while on treatment. FACT-P total score is the combined score (range 0–156) from 27 core items that assess subject function in four domains: physical, social/family, emotional, and functional well-being, in addition to 12 site-specific

	items which assess for prostate-related symptoms, each rated on a 0 to 4 Likert-type scale, in this instrument designed specifically for use with prostate cancer. Higher scores represent better QoL (range 0–156)[5] Continuous data was described by the number of observations, number of missing observations, mean, standard deviation, median, minimum, and maximum. Statistical comparisons were made using a two-sided test at the 0.05 significance level
Time to progression of worst pain	Time from randomization to an increase of ≥ 2 points in worst pain (item 3) from baseline[6] using the BPI-SF, a validated instrument that is a subject self-rating scale (ranging from 0 to 10, with higher scores indicating higher degree of pain) assessing level of pain, effect of the pain on activities of daily living, and analgesic use The analysis method was the same as the primary analysis for rPFS
Time to progression of pain severity	Time from randomization to an increase of ≥ 2 points in pain severity score (items 3–6) from baseline[6] using the BPI-SF, a validated instrument that is a subject self-rating scale (ranging from 0 to 10, with higher scores indicating higher degree of pain) assessing level of pain, effect of the pain on activities of daily living, and analgesic use The analysis method was the same as the primary analysis for rPFS

Abbreviations: BPI-SF, Brief Pain Inventory–Short Form; CI, confidence interval; FACT-P, Functional Assessment of Cancer Therapy–Prostate; HR, hazard ratio; ORR, objective response rate; OS, overall survival; PCWG2, Prostate Cancer Working Group 2 criteria; PRO, patient-reported outcomes; PSA, prostate-specific antigen; QLQ-PR25, Quality of Life PRostate-specific questionnaire; QoL, quality of life; RECIST, version 1.1, Response Evaluation Criteria in Solid Tumors, version 1.1; rPFS, radiographic progression-free survival; SSE, symptomatic skeletal event.

Table A2. Radiographic Progression-Free Survival Definition per RECIST, Version 1.1,[2] and Adapted from PCWG2.[3]

Date progression detected (visit)*	Criteria for progression	Criteria for confirmation of progression (requirement and timing)	Criteria for documentation of disease progression on confirmatory scan
Week 13			
Bone lesions	≥ 2 new lesions compared to baseline bone scan	Timing: ≥ 6 weeks after progression identified or at week 25 visit	≥ 2 new bone lesions on bone scan compared to week 13 scan (≥ 4 new lesions compared to baseline bone scan)
Soft tissue lesions	Progressive disease on CT or MRI by RECIST, version 1.1	No confirmatory scan required for soft tissue disease progression	Not applicable
Week 25 or later			
Bone lesions	≥ 2 new lesions on bone scan compared to best response on treatment	No confirmatory scan required	Not applicable
Soft tissue lesions	Progressive disease on CT or MRI by RECIST, version 1.1	No confirmatory scan required for soft tissue disease progression	Not applicable

*Progression detected by bone scan at an unscheduled visit prior to week 25 will require the same criteria for documentation of disease progression as week 13 with a confirmatory scan at least 6 weeks later or at the next scheduled scan.

Abbreviations: CT, computed tomography; MRI, magnetic resonance imaging; PCWG2, Prostate Cancer Working Group 2 Criteria; RECIST, version 1.1, Response Evaluation Criteria in Solid Tumors, version 1.1.

Table A3. First New Antineoplastic Prostate Cancer Therapy.

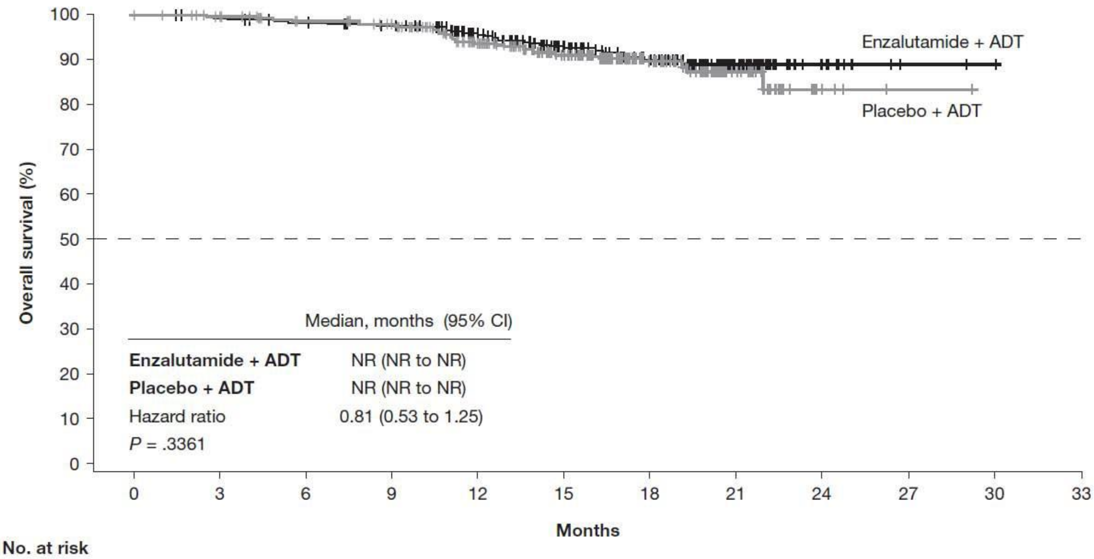
n (%)	Enzalutamide plus ADT (n = 574)	Placebo plus ADT (n = 576)
Overall	46 (8.0)	133* (23.1)
Most common in ≥ 2% of patients in either treatment group†		
Docetaxel	11 (23.9)	52 (39.1)
Abiraterone	13 (28.3)	28 (21.1)
Enzalutamide	4 (8.7)	28 (21.1)
Bicalutamide	4 (8.7)	12 (9.0)
Other	14 (30.4)	15 (11.3)

*One patient in the placebo plus ADT group received a combination of docetaxel with carboplatin and one patient in the placebo plus ADT group received a combination of docetaxel with blinded therapy. Both patients are counted in both the docetaxel and the other categories.

†Enzalutamide plus ADT, n = 46; placebo plus ADT, n = 133.

Abbreviation: ADT, androgen deprivation therapy.

Fig A2. Kaplan-Meier Estimate of Overall Survival (Intent-To-Treat Population).

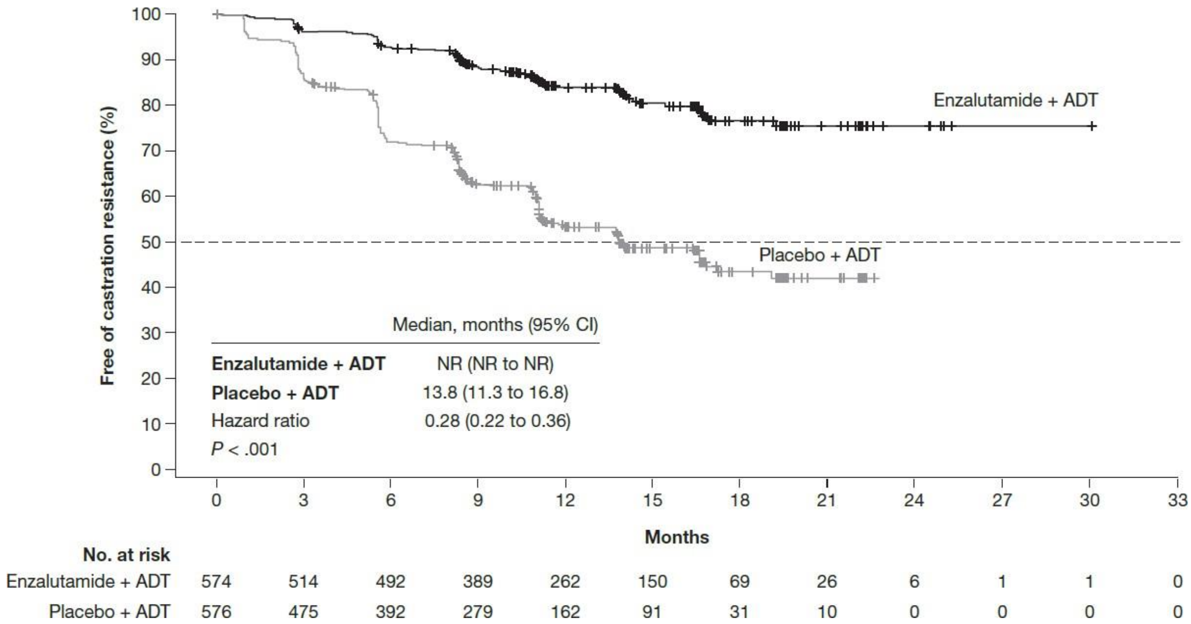


No. at risk												
Enzalutamide + ADT	574	568	559	548	379	258	130	45	13	2	1	0
Placebo + ADT	576	564	548	532	353	236	116	33	5	1	0	0

For patients still alive at the date of the analysis cut-off point, overall survival was censored on the last date the patient was known to be alive. The dashed line at the 50th percentile indicates the median. Crosses indicate censored data.

Abbreviations: ADT, androgen deprivation therapy; CI, confidence interval; NR, not reached.

Fig A3. Kaplan-Meier Estimate of Time to Castration Resistance (Intent-To-Treat Population).

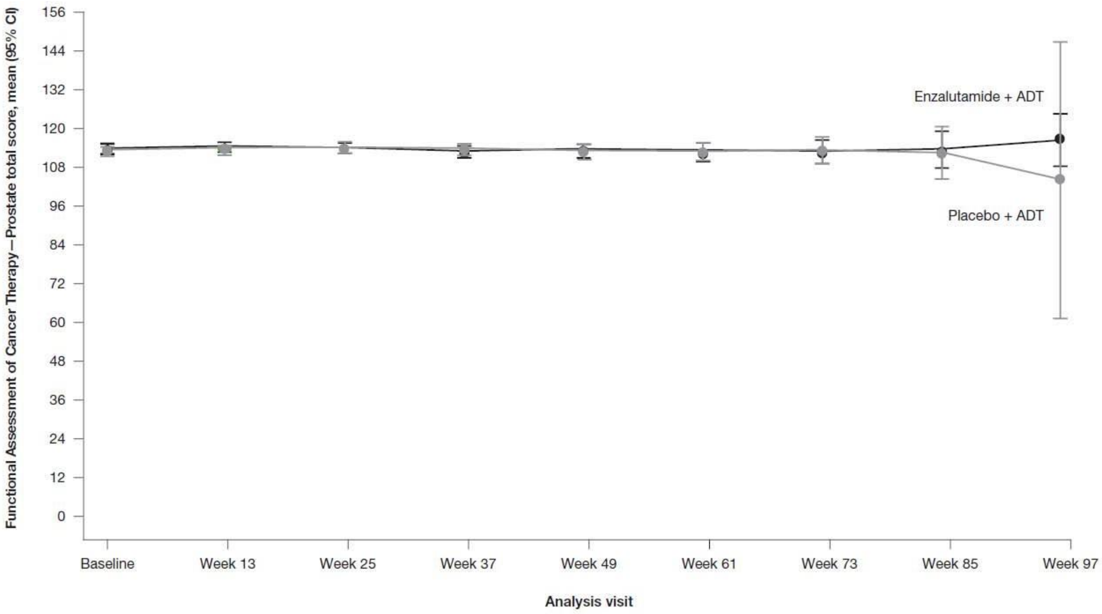


The dashed line at the 50th percentile indicates the median. Crosses indicate censored data.

In patients with no documented castration-resistance event, the time to castration resistance was censored on the latest date from: the date of last radiologic assessment, the last prostate-specific antigen sample taken prior to the start of any new prostate cancer therapy and prior to 2 or more consecutive missed prostate-specific antigen assessments (if applicable), and the last visit date performed.

Abbreviations: ADT, androgen deprivation therapy; CI, confidence interval; NR, not reached.

Fig A4. Mean Functional Assessment of Cancer Therapy–Prostate Total Score Over Time (Intent-To-Treat Population).

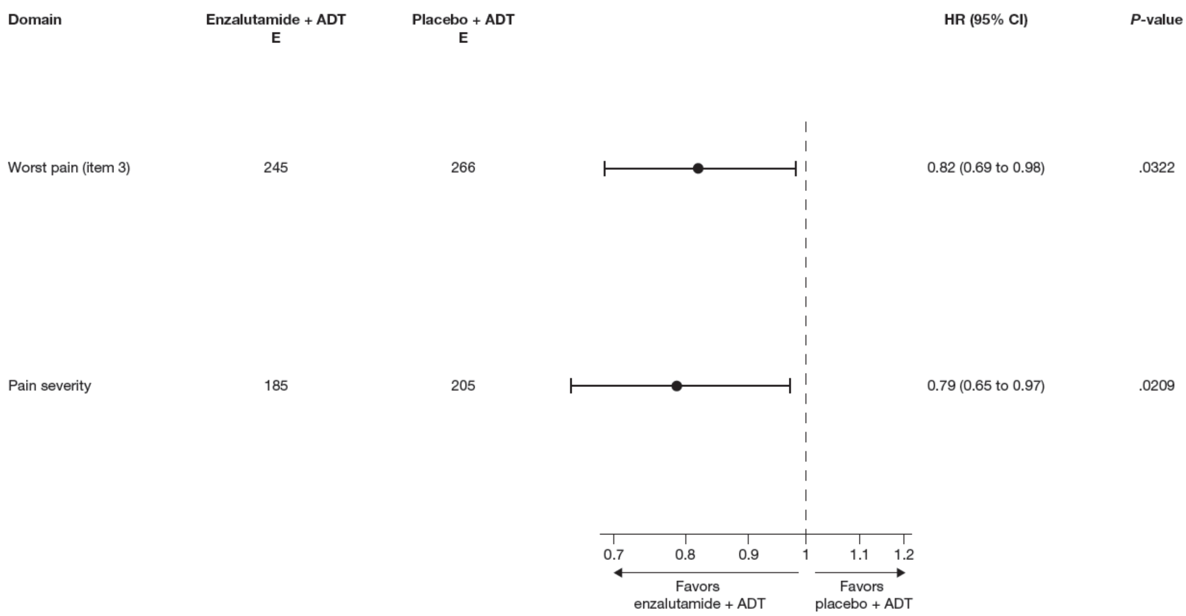


No. of patients									
Enzalutamide + ADT	550	533	499	474	349	236	128	51	18
Placebo + ADT	553	529	487	429	298	191	101	36	6

Visits up to week 97 only are presented due to small sample size beyond that timepoint.

Abbreviations: ADT, androgen deprivation therapy; CI, confidence interval.

Fig A5. Forest Plot of Prespecified Patient-Reported Outcomes Sensitivity Analyses of Time to Pain Progression (Intent-To-Treat Population).



Pain progression is defined as a ≥2-point increase in average Brief Pain Inventory–Short Form score for worst pain (item 3) and pain severity.

Abbreviations: ADT, androgen deprivation therapy; CI, confidence interval; E, number of events; HR, hazard ratio.

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